UNIT I

1 Introduction to Data Science and Big Data

Syllabus

Basics and need of Data Science and Big Data, Applications of Data Science, Data explosion, 5 V's of Big Data, Relationship between Data Science and Information Science, Business intelligence versus Data Science, Data Science Life Cycle, Data : Data Types, Data Collection. Need of Data wrangling, Methods : Data Cleaning, Data Integration, Data Reduction, Data Transformation, Data Discretization.

Contents

1.1 Basics and Need of Data Science and Big Data

- Data is collection of facts and figures which relay something specific, but which are not organized in any way. It can be numbers, words, measurements, observations or even just descriptions of things. We can say, data is raw material in the production of information.
- Types of data are record data, data matrix, document data, transaction data, graph data and ordered data.

Data science :

- Data science is an interdisciplinary field that seeks to extract knowledge or insights from various forms of data. At its core, data science aims to discover and extract actionable knowledge from data that can be used to make sound business decisions and predictions.
- Data science uses advanced analytical theory and various methods such as time series analysis for predicting future. From historical data, instead of knowing how many products sold in previous quarter, data science helps in forecasting future product sales and revenue more accurately.
- Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information and make business decisions. Data science uses complex machine learning algorithms to build predictive models.
- Data science enables businesses to process huge amounts of structured and unstructured big data to detect patterns.

Big data :

- Big data can be defined as very large volumes of data available at various sources, in varying degrees of complexity, generated at different speed i.e., velocities and varying degrees of ambiguity, which cannot be processed using traditional technologies, processing methods, algorithms or any commercial off-the-shelf solutions.
- 'Big data' is a term used to describe collection of data that is huge in size and yet growing exponentially with time. In short, such a data is so large and complex that none of the traditional data management tools are able to store it or process it efficiently.
- The processing of big data begins with the raw data that isn't aggregated or organized and is most often impossible to store in the memory of a single computer.

1.1.1 Difference between Data Science and Big Data

Sr. no.	Data science	Big data
1.	It is a field of scientific analysis of data in order to solve analytically complex problems and the significant and necessary activity of cleansing, preparing of data.	Big data is storing and processing large volume of structured and unstructured data that can not be possible with traditional applications.
2.	It is used in Biotech, energy, gaming and insurance.	Used in retail, education, healthcare and social media.
3.	Goals : Data classification, anomaly detection, prediction, scoring and ranking.	Goals : To provide better customer service, identifying new revenue opportunities, effective marketing etc.

1.1.2 Applications of Data Science

- Asking a personal assistant like Alexa or Siri for a recommendation demands data science. So does operating a self - driving car, using a search engine that provides useful results or talking to a chatbot for customer service. These are all real - life applications for data science.
- Following are some main reasons for using data science technology :
  - With the help of data science technology, we can convert the massive amount of raw and unstructured data into meaningful insights.
  - Data science technology is opting by various companies, whether it is a big brand or a startup. Google, Amazon, Netflix, which handle the huge amount of data are using data science algorithms for better customer experience.
  - Data science is working for automating transportation such as creating a self - driving car, which is the future of transportation.
- Data science can help in different predictions such as various survey, elections, flight ticket confirmation, etc.
  1. **Healthcare :** Healthcare companies are using data science to build sophisticated medical instruments to detect and cure diseases.
  2. **Gaming :** Video and computer games are now being created with the help of data science and that has taken the gaming experience to the next level.
  3. **Image recognition :** Identifying patterns in images and detecting objects in an image is one of the most popular data science applications.
  4. **Logistics :** Data science is used by logistics companies to optimize routes to ensure faster delivery of products and increase operational efficiency.

5. **Predict future market trends :** Collecting and analysing data on a larger scale can enable to identify emerging trends in market. Tracking purchase data, celebrities and influencers and search engine queries can reveal what products people are interested in.

6. **Recommendation systems :** Netflix and Amazon give movie and product recommendations based on what you like to watch, purchase or browse on their platforms.

7. **Streamline manufacturing :** Another way we can use data science in business is to identify inefficiencies in manufacturing processes. Manufacturing machines gather data from production processes at high volumes. In cases where the volume of data collected is too high for a human to manually analyse it, an algorithm can be written to clean, sort and interpret it quickly and accurately to gather insights.

1.2 Data Explosion

- The essence of computer applications is to store things in the real world into computer systems in the form of data, i.e., it is a process of producing data. Some data are the records related to culture and society and others are the descriptions of phenomena of universe and life. The large scale of data is rapidly generated and stored in computer systems, which is called data explosion.
- Data is generated automatically by mobile devices and computers, think Facebook, search queries, directions and GPS locations and image capture.
- Sensors also generate volumes of data, including medical data and commerce location - based sensors. Experts expect 55 billion IP - enabled sensors by 2021. Even storage of all this data is expensive. Analysis gets more important and more expensive every year.
- Fig. 1.2.1 shows the big data explosion by the current data boom and how critical it is for us to be able to extract meaning from all of this data.

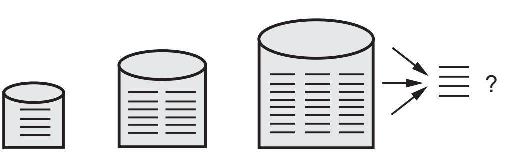

Fig. 1.2.1 Data explosion

- The phenomena of exponential multiplication of data that gets stored is termed as "Data Explosion". Continuous inflow of real - time data from various processes, machinery and manual inputs keeps flooding the storage servers every second.

- Sending emails, making phone calls, collecting information for campaigns; each day we create a massive amount of data just by going about our normal business and this data explosion does not seem to be slowing down. In fact, 90 % of the data that currently exists was created in just the last two years.
- Reason for this data explosion is **Innovation**.

1. **Business model transformation :** Innovation changed the way in which we do business, provide services. The data world is governed by three fundamental trends are business model transformation, globalization and personalization of services.
   - Organizations have traditionally treated data as a legal or compliance requirement, supporting limited management reporting requirements. Consequently, organizations have treated data as a cost to be minimized.
   - The businesses are required to produce more data related to product and provide services to cater each sector and channel of customer.
2. **Globalization :** Globalization is an emerging trend in business where organizations start operating on international scale. From manufacturing to customer service, globalization have changed the commerce of the world. Variety and different formats of data is generated due to globalization.
3. **Personalization of services :** To enhance customer service, the form of one - to - one marketing in the form of personalization of service is opted by customer. Customer expects communication through various channels increases the speed of data generation.
4. **New sources of data :** The shift to online advertising supported by the likes of Google, Yahoo, and others is a key driver in the data boom. Social media, mobile devices, sensor networks and new media are on the fingertips of customer or user. The data generated through this is used by corporations for decision support systems like Business Intelligence and analytics. The growth of technology helped to emerge new business models over the last decade or more. Integration of all the data across the enterprise is used to create business decision support platform.

1.3 5 V's of Big Data

- We differentiate big data characteristics from traditional data by one or more of the five V's : Volume, velocity, variety, veracity and value.

1. **Volume :** Volumes of data are larger than conventional relational database infrastructure can cope with. It consists of terabytes or petabytes of data.

- Fig. 1.3.1 shows big data volume.

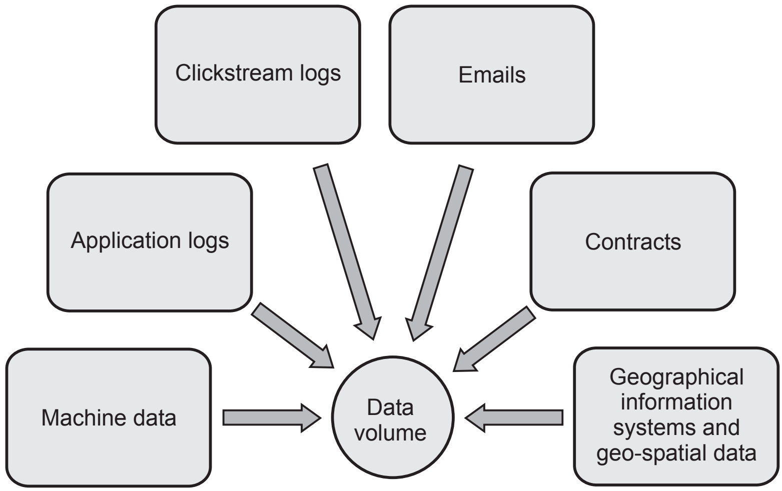

Fig. 1.3.1 Big data volume

2) **Velocity :** The term 'velocity' refers to the speed of generation of data. How fast the data is generated and processed to meet the demands, determines real potential in the data. It is being created in or near real - time.

3) **Variety :** It refers to heterogeneous sources and the nature of data, both structured and unstructured.

- Fig. 1.3.2 shows big data velocity and data variety.

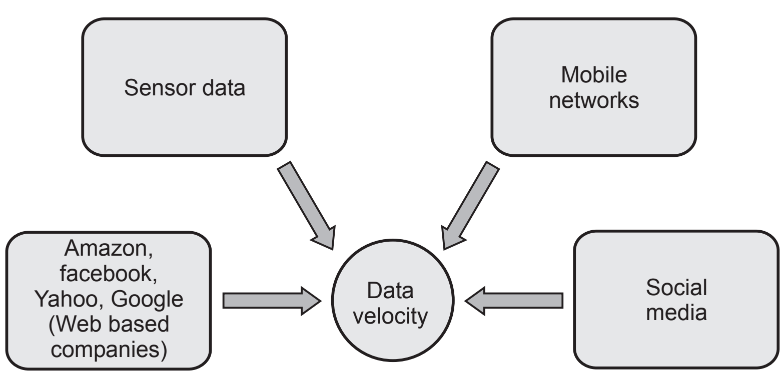

Fig. 1.3.2 (a)

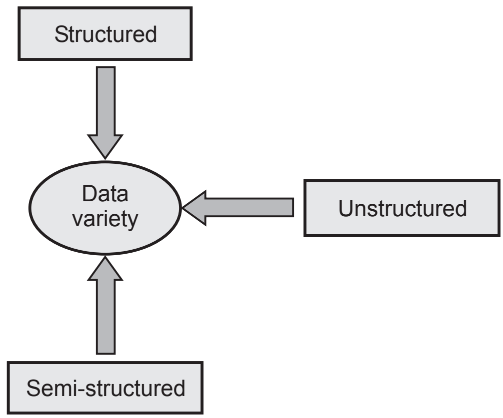

Fig. 1.3.2 (b)

4. **Value :** It represents the business value to be derived from big data.
   - The ultimate objective of any big data project should be to generate some sort of value for the company doing all the analysis.
   - For real - time spatial big data, decisions can be enhance through visualization of dynamic change in such spatial phenomena as climate, traffic, social - media - based attitudes and massive inventory locations.
   - Exploration of data trends can include spatial proximities and relationships. Once spatial big data are structured, formal spatial analytics can be applied, such as spatial autocorrelation, overlays, buffering, spatial cluster techniques and location quotients.
5. **Veracity :** Big data must be fed with relevant and true data. We will not be able to perform useful analytics if many of the incoming data comes from false sources or has errors. Veracity refers to the level of trustiness or messiness of data and if higher the trustiness of the data, then lower the messiness and vice versa. It relates to the assurance of the data's quality, integrity, credibility and accuracy. We must evaluate the data for accuracy before using it for business insights because it is obtained from multiple sources.

1.4 Relationship between Data Science and Information Science

SPPU : Dec.-18

- Data science, as the interdisciplinary field, employs techniques and theories drawn from many fields within the context of mathematics, statistics, information science and computer science. Data science and information science are twin disciplines by nature. The mission, task and nature of data science are consistent with those of information science.

- Data science is heavy on computer science and mathematics. Information science is used in areas such as knowledge management, data management and interaction design.
- Information science is the science and practice dealing with the effective collection, storage, retrieval and use of information. It is concerned with recordable information and knowledge and the technologies and related services that facilitate their management and use.

1.4.1 Business Intelligence versus Data Science

Business Intelligent (BI)	Data Science
Bl tends to provide reports, dashboards, and queries on business questions for the current period or in the past.	Data science tends to use disaggregated data in a more forward-looking, exploratory way, focusing on analyzing the present and enabling informed decisions about the future.
Bl systems make it easy to answer questions related to quarter-to-date revenue, progress toward quarterly targets, and understand how much of a given product was sold in a prior quarter or year.	Data science tends to be more exploratory in nature and may use scenario optimization to deal with more open - ended questions.
BI helps monitor the current state of business data to understand the historical performance of a business.	Data science, as used in business, is basically data-driven, where many interdisciplinary sciences are applied together to extract meaning.
BI is designed to handle static and highly structured data.	Data science can handle high-speed, high-volume and complex, multi-structured data from a wide variety of data sources.

1.4.2 Compare Cloud Computing and Big Data

Sr. No.	Cloud computing	Big data
1.	It provides resources on demand.	It provides a way to handle huge volumes of data and generate insights.
2.	It refers to internet services from SaaS, PaaS to IaaS.	It refers to data, which can be structured, semi-structured or unstructured.
3.	Cloud is used to store data and information on remote servers.	It is used to describe huge volume of data and information.
4.	Cloud computing is economical as it has low maintenance costs centralized platform no upfront cost and disaster safe implementation.	Big data is highly scalable, robust ecosystem and cost - effective.

5.	Vendors and solution providers of cloud computing are Google, Amazon Web Service, Dell, Microsoft, Apple and IBM.	Vendors and solution providers of big data are Cloudera, Hortonworks, Apache and MapR.
6.	The main focus of cloud computing is to provide computer resources and services with the help of network connection.	Main focus of big data is about solving problems when a huge amount of data generating and processing.

Review Question

1. *Compare BI Vs. data science.* **SPPU : Dec.-18 (End Sem), Marks 6**

1.5 Data Science Life Cycle

- A data science life cycle is an iterative set of data science steps you take to deliver a project or analysis. Fig 1.5.1 shows data science life cycle.

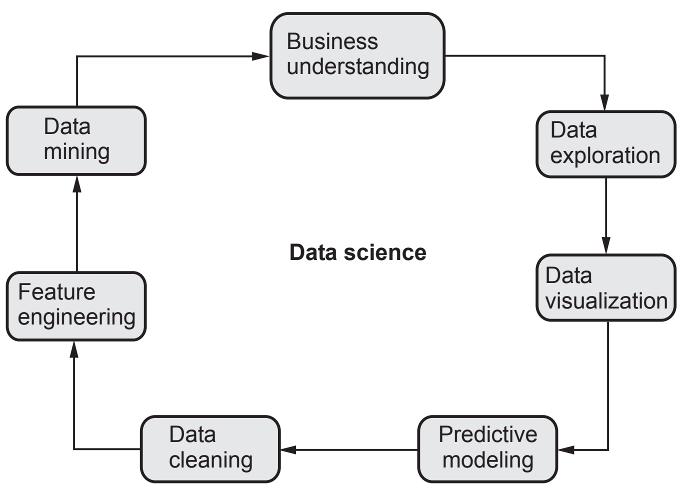

Fig. 1.5.1 Data science life cycle

a) **Business understanding :** Understand the basic problem you are trying to solve.

b) **Data exploration :** Understand the pattern and bias in your data.

c) **Data visualization :** Create and study of the visual representation of data.

d) **Predictive modeling :** It is the stage where the machine learning finally comes into data.

e) **Data cleaning :** Detecting and correcting corrupt or inaccurate records.

f) **Feature engineering :** It is the process of cutting down the features.

g) **Data mining :** Gathering your data from different source.

1.6 Data

- Data is collection of facts and figures which relay something specific, but which are not organized in any way. It can be numbers, words, measurements, observations or even just descriptions of things. We can say, data is raw material in the production of information.
- **Examples of data :** Printed paper, bank account passbook, student attendance, salary sheet etc.
- Data creation is limited because of lack of new technology. After start of using computer, data can be converted into more convenient forms. It starts of using email, e-book, digital audio and video, images etc.
- Data is raw. It simply exists and has no significance beyond its existence. It can exist in any form, usable or not. It does not have meaning of itself. In computer parlance, a spreadsheet generally starts out by holding data.
- Facts and figures which relay something specific, but which are not organized in any way and which provide no further information regarding patterns, context, etc.
- Data represents a fact or statement of event without relation to other things. Example : It is raining. Computer stores data in the form of 0 and 1. This is called digital data. These data are stored in the computer in the form of 0 and 1. Data in this form is called digital data. It is shown in Fig. 1.6.1.

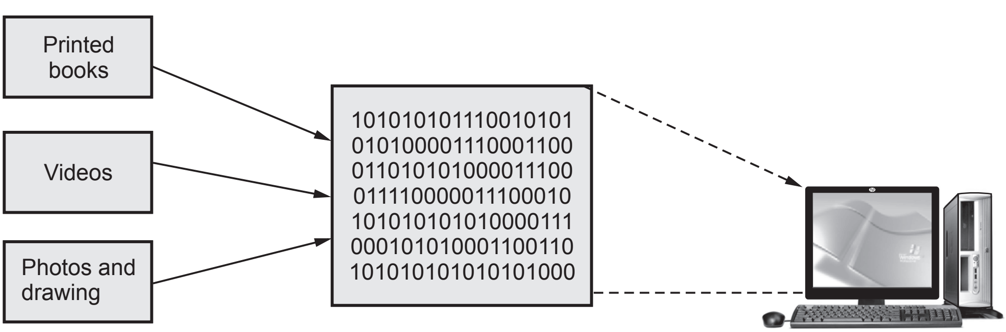

Fig. 1.6.1 Digital data

- Data is raw. It has not been shaped, processed or interpreted. Once data has been processed and turned into information. Computers need data. Humans need information. Data is a building block. Information gives meaning and context.
- Information systems are combinations of hardware, software and telecommunications networks that people build and use to collect, create and distribute useful data, typically in organizational settings.

- Company offers their products to customers for making money. These products can be goods or services. Large number of different types of data is accumulated in the organization. It contains data of products, customer data, employee's data, data of the delivery of products and data of other sources.
- These data must be stored, managed and processed. For performing these operations on data, information system can play an important role. Because this data is raw data. There is no unique format.
- Because of new technology, generation of new data and sharing of data has increased exponentially. The factors that affects the growth of digital data :
  1. Cost of digital storage device is decreases.
  2. Data processing capabilities is also increases.
  3. Faster access of data

1.6.1 Data Types

- Data are of two types : Structured and unstructured.

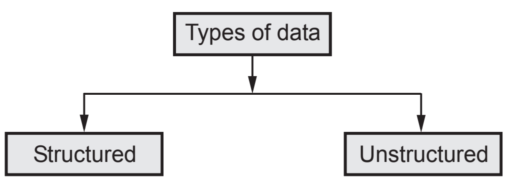

Fig. 1.6.2

Structured data :

- Structured data is arranged in rows and column format. It helps for application to retrieve and process data easily. Database management system is used for storing structured data.
- The term structured data refers to data that is identifiable because it is organized in a structure. The most common form of structured data or records is a database where specific information is stored based on a methodology of columns and rows.
- Structured data is also searchable by data type within content. Structured data is understood by computers and is also efficiently organized for human readers.

Unstructured data

- Unstructured data is data that does not follow a specified format. Row and columns are not used for unstructured data. Therefore it is difficult to retrieve required information. Unstructured data has no identifiable structure.

- The unstructured data can be in the form of text : (Documents, email messages, customer feedbacks), audio, video, images. Email is an example of unstructured data.
- Even today in most of the organizations more than 80 % of the data are in unstructured form. This carries lots of information. But extracting information from these various sources is a very big challenge.
- Characteristics of unstructured data :
  1. There is no a structural restriction or binding for the data.
  2. Data can be of any type.
  3. Unstructured data does not follow any structural rules.
  4. There are no predefined formats, restriction or sequence for unstructured data.
  5. Since there is no structural binding for unstructured data it is unpredictable in nature.

Examples of machine generated unstructured data :

1. **Satellite images :** This includes weather data or the data that the government captures in its satellite surveillance imagery.
2. **Scientific data :** This includes atmospheric data and high energy physics.
3. **Photographs and video :** This include security, surveillance and traffic video.

1.6.2 Difference between Structured and Unstructured Data

Parameters	Structured data	Unstructured data
Representation	It is in discrete form. i.e. stored in row and column format.	Unstructured data is data that does not follow a specified format.
Metadata	Syntax	Semantics
Storage	Database management system	Unmanaged file structure
Standard	SQL, ADO.net, ODBC	Open XML, SMTO, SMS
Tools for integration	ETL	Batch processing or manual data entry.
characteristics	With a structured document, certain information always appears in the same location on the page.	In unstructured document information can appear in unexpected places on the document.
Used by organizations	Low volume operations	High volume operations

1.6.3 Difference between Information and Data

Sr. No.	Information	Data
1.	It is processed data.	It is raw data.
2.	Information is specific.	Data is not specific.
3.	Information depends on data.	Data does not depend on information.
4.	Information is output of computer.	Data is input to the computer.
5.	Information simply refers to the knowledge of value obtained through the collection, interpretation and/or analysis of data.	Data refer to facts, measurements, characteristics, or traits of an object of interest.

1.6.4 Qualitative and Quantitative Data

- Data can broadly be divided into following two types :
  1. Qualitative data
  2. Quantitative data

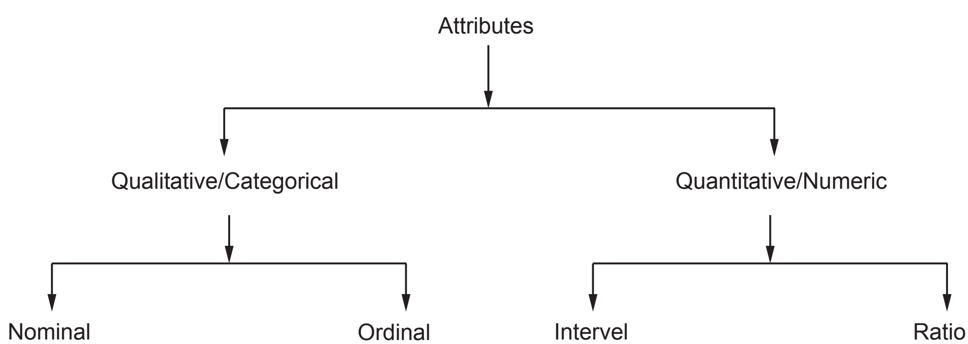

Fig. 1.6.3

Qualitative data :

- **Qualitative data** provides information about the quality of an object or information which cannot be measured. Qualitative data cannot be expressed as a number. Data that represent nominal scales such as gender, economic status, religious preference are usually considered to be qualitative data.
- **Qualitative data** is data concerned with descriptions, which can be observed but cannot be computed. Qualitative data is also called categorical data. Qualitative data can be further subdivided into two types as follows :
  1. Nominal data
  2. Ordinal data

Nominal data

- A nominal data is the 1^{st} level of measurement scale in which the numbers serve as "tags" or "labels" to classify or identify the objects.
- A nominal data usually deals with the non-numeric variables or the numbers that do not have any value. While developing statistical models, nominal data are usually transformed before building the model.
- It is also known as categorical variables.

Characteristics of nominal data :

1. A nominal data variable is classified into two or more categories. In this measurement mechanism, the answer should fall into either of the classes.
2. It is qualitative. The numbers are used here to identify the objects.
3. The numbers don't define the object characteristics. The only permissible aspect of numbers in the nominal scale is "counting."

- Example :

1. Gender : Male, Female, Other.
2. Hair color : Brown, Black, Blonde, Red, Other.

Ordinal data

- Ordinal data is a variable in which the value of the data is captured from an ordered set, which is recorded in the order of magnitude.
- Ordinal represents the "order." Ordinal data is known as qualitative data or categorical data. It can be grouped, named and also ranked.
- Characteristics of the ordinal data :

a) The ordinal data shows the relative ranking of the variables.

b) It identifies and describes the magnitude of a variable.

c) Along with the information provided by the nominal scale, ordinal scales give the rankings of those variables.

d) The interval properties are not known.

e) The surveyors can quickly analyze the degree of agreement concerning the identified order of variables.

- Examples :

a) University ranking : 1^{st}, 9^{th}, 87^{th}…

b) Socioeconomic status : Poor, middle class, rich.

c) Level of agreement : Yes, maybe, no.

d) Time of day : Dawn, morning, noon, afternoon, evening, night.

Quantitative data

- **Quantitative data** is the one that focuses on numbers and mathematical calculations and can be calculated and computed.
- **Quantitative data** are anything that can be expressed as a number, or quantified. Examples of quantitative data are scores on achievement tests, number of hours of study, or weight of a subject. These data may be represented by ordinal, interval or ratio scales and lend themselves to most statistical manipulation.
- There are two types of quantitative data : Interval data and Ratio data

Interval data :

- Interval data corresponds to a variable in which the value is chosen from an interval set.
- It is defined as a quantitative measurement scale in which the difference between the two variables is meaningful. In other words, the variables are measured in an exact manner, not as in a relative way in which the presence of zero is arbitrary.
- Characteristics of interval data :
  a) The interval data is quantitative as it can quantify the difference between the values.
  b) It allows calculating the mean and median of the variables.
  c) To understand the difference between the variables, you can subtract the values between the variables.
  d) The interval scale is the preferred scale in statistics as it helps to assign any numerical values to arbitrary assessment such as feelings, calendar types, etc.
- Examples :
  1. Celsius temperature.
  2. Fahrenheit temperature.
  3. Time on a clock with hands.

Ratio data :

- Any variable for which the ratios can be computed and are meaningful is called ratio data.
- It is a type of variable measurement scale. It allows researchers to compare the differences or intervals. The ratio scale has a unique feature. It possesses the character of the origin or zero points.
- Characteristics of ratio data :
  a) Ratio scale has a feature of absolute zero.
  b) It doesn't have negative numbers, because of its zero - point feature.

c) It affords unique opportunities for statistical analysis. The variables can be orderly added, subtracted, multiplied, divided. Mean, median, and mode can be calculated using the ratio scale.

d) Ratio data has unique and useful properties. One such feature is that it allows unit conversions like kilogram - calories, gram - calories, etc.

- Examples : Age, Weight, Height, Ruler measurements, Number of children

1.6.5 Difference between Qualitative and Quantitative Data

Qualitative data	Quantitative data
Qualitative data provides information about the quality of an object or information which cannot be measured	Quantitative data relates to information about the quantity of an object; hence it can be measured
Types : Nominal data and Ordinal data	Types : Interval data and Ratio data
Narratives often make use of adjectives and other descriptive words to refer to data on appearance, color, texture, and other qualities	Measure's quantities such as length, size, amount, price, and even duration.
They are descriptive rather than numerical in nature	Expressed in numerical form.
For example : • The team is well prepared. • The leaf feels waxy. • The river is peaceful.	For example : • The team has 7 players. • The leaf weighs 2 ounces. • The river is 25 miles long.

1.6.6 Data Collection

- Data collection is the systematic approach to gathering and measuring information from a variety of sources to get a complete and accurate picture of an area of interest. The big data includes information produced by humans and devices.
- The big data collection is focused on the following types of data :

  a) Network data : This type of data is gathered on all kinds of networks, including social media, information and technological networks, the Internet and mobile networks, etc.

  b) Real - time data : They are produced on online streaming media, such as YouTube, Twitch, Skype or Netflix.

  c) Transactional data : They are gathered when a user makes an online purchase (information on the product, time of purchase, payment methods, etc.).

d) **Geographic data :** Location data of everything, humans, vehicles, building, natural reserves and other objects are continuously supplied with satellites.

e) **Natural language data :** These data are gathered mostly from voice searches that can be made on different devices accessing the Internet.

f) **Time series data :** This type of data is related to the observation of trends and phenomena taking place at this very moment and over a period of time, for instance, global temperatures, mortality rates, pollution levels, etc.

1.7 Data Wrangling

- Data wrangling is the process of getting the data from its raw format into something suitable for more conventional analytics. Data wrangling can be defined as the process of cleaning, organizing and transforming raw data into the desired format for analysts to use for prompt decision - making.
- The data is often stored in a special purpose database that requires specialized tools to access. Fig 1.7.1 shows data wrangling process.

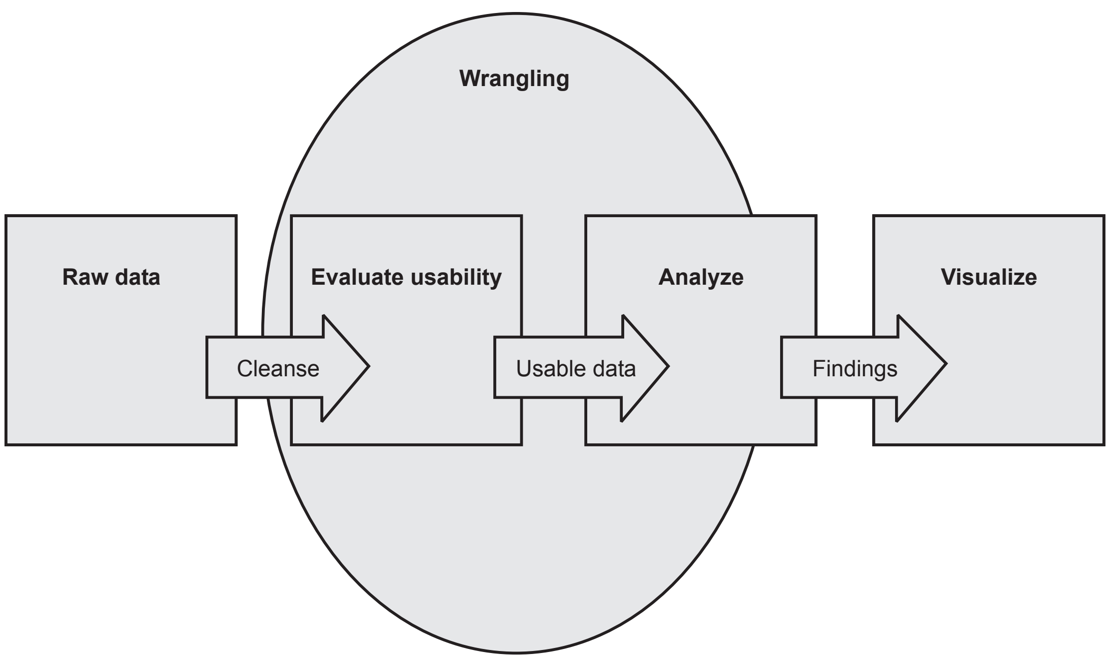

Fig. 1.7.1 Data wrangling process

- Data wrangling is the process of cleaning, structuring and enriching raw data into a desired format for better decision making in less time.
- The **necessity for data wrangling** is often a by - product of poorly collected or presented data. Data that is entered manually by humans is typically fraught with errors; data collected from websites is often optimized to be displayed on websites, not to be sorted and aggregated.

- Tasks in data wrangling include :
  a) Merging multiple datasets into one large dataset for analysis.
  b) Examining missing/gaps in data.
  c) Removing outliers or anomalies in datasets.
  d) Standardizing inputs.
- **Goals of data wrangling :**
  a) Creating valid and novel data out of messy data to drive decision - making in businesses.
  b) Standardizing raw data into formats that big data systems can ingest.
  c) Reducing the time spent by data analysts when creating data models by presenting orderly data.
  d) Creating consistency, completeness, usability and security for any dataset consumed or stored in a data warehouse.
- There are typically six iterative steps that make up the data wrangling process :
  1. **Discovering :** Before you can dive deeply, you must better understand what is in your data, which will inform how you want to analyze it. How you wrangle customer data, for example, may be informed by where they are located, what they bought or what promotions they received.
  2. **Structuring :** This means organizing the data, which is necessary because raw data comes in many different shapes and sizes. A single column may turn into several rows for easier analysis. One column may become two. Movement of data is made for easier computation and analysis.
  3. **Cleaning :** What happens when errors and outliers skew user data ? User clean the data. What happens when state data is entered as AP or Andhra Pradesh or Arunachal Pradesh ? You clean the data. Null values are changed and standard formatting implemented, ultimately increasing data quality.
  4. **Enriching :** Here you take stock in data and strategize about how other additional data might augment it. Questions asked during this data wrangling step might be : What new types of data can derive from what already have or what other information would better inform my decision making about this current data ?
  5. **Validating :** Validation rules are repetitive programming sequences that verify data consistency, quality and security. Examples of validation include ensuring uniform distribution of attributes that should be distributed normally (e.g., birth dates) or confirming accuracy of fields through a check across data.

6. **Publishing :** Analysts prepare the wrangled data for use downstream - Whether by a particular user or software and document any particular steps taken or logic used to wrangle said data. Data wrangling gurus understand that implementation of insights relies upon the ease with which it can be accessed and utilized by others.

1.7.1 Benefits of Data Wrangling

a) Data wrangling helps to improve data usability as it converts data into a compatible format for the end system.

b) It helps to quickly build data flows.

c) Integrates various types of information and their sources.

d) Help users to process very large volumes of data easily and easily share data - flow techniques.

e) Improved efficiency when it comes to data - driven decision making.

1.8 Data Cleaning

- Sometimes real - world data is incomplete, noisy, and inconsistent. Data cleaning methods are used for making useable data.
- Data cleaning tasks are as follows :
  1. Data acquisition and metadata
  2. Fill in missing values
  3. Unified date format
  4. Converting nominal to numeric
  5. Identify outliers and smooth out noisy data
  6. Correct inconsistent data
- Data cleaning is a first step in data pre-processing techniques which is used to find the missing value, smooth noise data, recognize outliers and correct inconsistent.

1.8.1 Missing Value

- These dirty data will affects on miming procedure and led to unreliable and poor output. Therefore it is important for some data cleaning routines.

How to handle noisy data in data mining ?

- Following methods are used for handling noisy data :
  1. **Ignore the tuple :** Usually done when the class label is missing. This method is not good unless the tuple contains several attributes with missing values.
  2. **Fill in the missing value manually :** It is time - consuming and not suitable for a large data set with many missing values.

3. **Use a global constant to fill in the missing value :** Replace all missing attribute values by the same constant.
4. **Use the attribute mean to fill in the missing value :** For example, suppose that the average salary of staff is ₹ 65000/-. Use this value to replace the missing value for salary.
5. Use the attribute mean for all samples belonging to the same class as the given tuple.
6. Use the most probable value to fill in the missing value.

1.8.2 Noisy Data

- **Noise :** Random error or variance in a measured variable.
- For numeric values, box plots and scatter plots can be used to identify outliers. To deal with these anomalous values, data smoothing techniques are applied, which are described below.

1. **Binning :** Using binning methods smooths sorted value by using the values around it. The sorted values are then divided into 'bins'. There are various approaches to binning. Two of them are smoothing by bin means where each bin is replaced by the mean of bin's values, and smoothing by bin medians where each bin is replaced by the median of bin's values.

Binning methods for data smoothing :

a) **In smoothing by bin means :** Each value in a bin is replaced by the mean value of the bin. For example, the mean of the values 5, 9 and 13 in Bin is 9. Therefore, each original value in this bin is replaced by the value 9.

b) **Smoothing by bin medians** can be employed, in which each bin value is replaced by the bin median.

c) **Smoothing by bin boundaries :** The minimum and maximum bin values are stored at the boundary while intermediate bin values are replaced by the boundary value to which it is more closer.

2. **Regression :** Linear regression and multiple linear regression can be used to smooth the data, where the values are conformed to a function.
3. **Outlier analysis :** Approaches such as clustering can be used to detect outliers and deal with them.

1.9 Data Integration and Transformation

1.9.1 Data Integration

- Data integration combines data from multiple sources to form a coherent data store. Metadata, correlation analysis, data conflict detection, and the resolution of semantic heterogeneity contribute toward smooth data integration.
- With the increasing volume of data collected through a variety of sources and at a much faster velocity every day, it is very much clear that Data is and has been the most valuable possession.
- Data integration is important as it provides a unified view of the scattered data not only this it also maintains the accuracy of data.
- Issues in data integration : While integrating the data we have to deal with several issues.

1. Entity identification problem

- As we know the data is unified from the heterogeneous sources then how can we 'match the real-world entities from the data'. For example, we have customer data from two different data source.
- An entity from one data source has **customer_id** and the entity from the other data source has **customer_number**. Now how does the data analyst or the system would understand that these two entities refer to the same attribute ? The schema integration can be achieved using metadata of each attribute.

2. Redundancy

- Redundancy is one of the big issues during data integration. Redundant data is an unimportant data or the data that is no longer needed. It can also arise due to attributes that could be derived using another attribute in the data set.
- For example, one data set has the customer age and other data set has the customers date of birth then age would be a redundant attribute as it could be derived using the date of birth.
- The redundancy can be discovered using correlation analysis. The attributes are analyzed to detect their interdependency on each other thereby detecting the correlation between them.

1.9.2 Data Transformation

- In data transformation, the data are transformed or consolidated into forms appropriate for mining.

- Data transformation can involve the following :
  1. **Smoothing :** It removes noise from the data. Such techniques include binning, regression and clustering.
  2. **Aggregation :** An aggregation or summary operation is applied to the data.
  3. **Generalization** of the data, where low-level or "primitive" (raw) data are replaced by higher-level concepts through the use of concept hierarchies.
  4. **Normalization :** The attribute data are scaled so as to fall within a small specified range.
  5. **Attribute construction :** New attributes are constructed and added from the given set of attributes to help the mining process.
- An attribute is normalized by scaling its values so that they fall within a small specified range. There are many methods for data normalization. They are min-max normalization, z-score normalization, and normalization by decimal scaling.
  a) Min-max normalization performs a linear transformation on the original data. It will scale the data between the 0 and 1.

Example :

Marks
8
10
15
20

Min : Minimum value of the given attribute. Here min is 8.

Max : Maxing value of the given attribute. Here max is 20.

V : V is the respective value of attribute.

For example :

V_1 : 8, V_2 = 10, V_3 = 15 and V_4 = 20.

New max : 1

Now min : 0

$$V' = \frac{V - Min_A}{Max_A - Min_A}(New - Max_A - New - Min_A) + New - Min_A$$

For mark 8 :

$$\text{minmax} = \frac{\text{V} - \text{min marks}}{\text{Max marks} - \text{Min marks}}(\text{New marks} - \text{New min}) + \text{New min}$$

$$\text{minmax} = \frac{8-8}{20-8} \times (1-0)+0$$

$$\text{minmax} = \frac{(0)}{12} \times 1$$

For mark 10 :

$$\text{minmax} = \frac{(10-8)}{20-8} \times (1-0)+0$$

$$\text{minmax} = \frac{2}{12} \times 1$$

$$\text{minmax} = 0.25$$

For mark 15 :

$$\text{minmax} = \frac{(15-8)}{20-8} \times (1-0)+0$$

$$\text{minmax} = \frac{(3)}{12} \times 1$$

$$\text{minmax} = 0.25$$

For mark 20 :

$$\text{minmax} = \frac{(20-8)}{20-8} \times (1-0)+0$$

$$\text{minmax} = \frac{12}{12} \times 1$$

$$\text{mimmax} = 1$$

Marks	Marks after min-max normalization
8	0
10	0.16
15	0.25
20	1

b) **Decimal scaling :** Decimal scaling is a data normalization technique. In this technique we move the decimal point of values of the attribute. His movement of decimal points totally depends on the maximum value among all values in the attribute.

Formula : A value V if attribute A can be obtained by normalization by the following formula.

Normalized value of attribute : = $(V^{i}/10^{j})$

Example :

CGPA	Formula	CGPA normalized after decimal scaling
2	2/10	0.2
3	3/10	0.3

We will check maximum value among our attribute CGPA. Here maximum value is 3 so we can convert it into decimal by dividing with 10.

Example 1.9.1 *1) Minimum salary is ₹ 20,000 and maximum salary is ₹ 1,70,000. Map the salary ₹ 1,00,000 in new range of ₹ (60,000, 2,60,000) using min-max normalization method.*

2) If mean salary is ₹ 54,000 and standard deviation is ₹ 16,000 then find z score value of ₹ 73,600 salary.

Solution :

Solution 1 :

Old range = (20000, 1,70,000)

max = 1,70,000

min = 20000

New range = (60000, 260000)

new_max = 260000

new_min = 60000

V_i = 100000

$$V_i' = [\{(V_i - \text{min})(\text{max} - \text{min})\} \times \{\text{new\_max} - \text{new\_min}\}] + \text{new\_min}$$

$$= [\{(80000/150000) \times 200000\}] + 60000$$

$$= [106666] + 60000$$

$$= 166666$$

Salary ₹ 100000 in old range is equal to salary ₹ 166666 in the new range.

Solution 2 :

$$\text{mean} = ₹\ 54{,}000$$

$$\text{Standard deviation} = ₹\ 16{,}000$$

$$\text{Z-score value of } 76{,}300 = \frac{(76{,}300 - \text{Mean})}{\text{Standard deviation}} = \frac{(54000)}{16000}$$

$$= \frac{54000}{16000} = 3.375$$

Z-score value of ₹ 73,600 salary is 3.375.

Example 1.9.2 *Use min-max normalization method to normalize the following group of data by setting min = 0 and max = 1, 200, 300, 400, 600, 1000.*

Solution :

i) Min-max normalization by setting min = 0 and max = 1.

Original data	200	300	400	600	1000
0, 1 normalized	0	0.125	0.25	0.5	1

ii) Z-score normalization

Original data	200	300	400	600	1000
0, 1 normalized	– 1.06	– 0.7	– 0.35	0.35	1.78

Example 1.9.3 *Suppose that the minimum and maximum values for the attribute income are \$ 73,600 and \$ 98,000, respectively. Normalize income value \$ 73,600 to the range [0:0 ; 1 : 0] using min-max normalization method.*

Solution : The min-max normalization to transform value 73,600 onto the range [0.0, 1.0].

Given data : $\text{min}_A = 12000$, $\text{max}_A = 98000$, $\text{new\_min}_A = 0.0$,

$$\text{new\_max}_A = 1.0,\ v = 73600,\ v' = ?$$

$$v' = \frac{v - \text{min}_A}{\text{max}_A - \text{min}_A} \times (\text{new\_max}_A - \text{new\_min}_A) + \text{new\_min}_A$$

$$v' = \frac{73600 - 12000}{98000 - 12000} \times (1.0 - 0.0) + 0.0$$

$$v' = 0.716$$

Example 1.9.4 *Consider the following group of data 200, 300, 400, 600, 1000.*

i) Use the min-max normalization to transform value 600 onto the range [0.0,1.0]

ii) Use the decimal scaling to transform value 600.

Solution : i) The min-max normalization to transform value 600 onto the range [0.0, 1.0].

Given data : $\min_A = 200$, $\max_A = 1000$, $new\_min_A = 0.0$,

$$new\_max_A = 1.0,\ v = 600,\ v' = ?$$

$$v' = \frac{v - \min_A}{\max_A - \min_A} \times (new\_max_A - new\_min_A) + new\_min_A$$

$$v' = \frac{600-200}{1000-200} \times (1.0-0.0)+0.0$$

$$v' = 0.5$$

ii) Decimal scaling to transform value 600

$$v = 600,\ j = 3$$

$$v = \frac{v}{10^j} = \frac{600}{10^3} = 0.6$$

1.10 Data Reduction

- Data reduction is nothing but obtaining a reduced representation of the data set that is much smaller in volume but yet produces the same analytical results.
- Data reduction is a process that reduced the volume of original data and represents it in a much smaller volume. Data reduction techniques ensure the integrity of data while reducing the data.
- Data reduction does not affect the result obtained from data mining that means the result obtained from data mining before data reduction and after data reduction is the same.
- Data reduction strategies are as follows :
  1. **Data cube aggregation :** Aggregation operations are applied to the data in the construction of a data cube.
  2. **Dimensionality reduction :** In dimensionality reduction redundant attributes are detected and removed which reduce the data set size.
  3. **Data compression :** Encoding mechanisms are used to reduce the data set size.
  4. **Numerosity reduction :** In numerosity reduction where the data are replaced or estimated by alternative.

5. **Discretisation and concept hierarchy generation :** Where raw data values for attributes are replaced by ranges or higher conceptual levels.

- Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. Mining on the reduced data set should be more efficient yet produce the same analytical results.
- Strategies for data reduction include the following :

1. **Data cube aggregation :** It is lowest level of a data cube. Summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to computer monthly and annual total amounts. This step is typically used in constructing a data cube for analysis of the data at multiple granularities.
2. **Attribute subset selection :** Attribute subset selection reduces the data set size by removing irrelevant or redundant attributes. The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.

- **Irrelevant attributes :** It contains no information that is useful for the data mining task at hand. For example; Student's roll number is often irrelevant to the task of predicting student marks or CGPA.
- **Redundant attributes :** Duplicate much or all of the information contained in one or more other attributes. For example : purchase price of a product and the amount of GST paid.

3. **Dimensionality reduction :** Dimensionality reduction is the process of reducing the number of random variables under consideration, by obtaining a set of principal variables.

- If the original data can be reconstructed from the compressed data without any loss of information, the data reduction is called **lossless**. If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called **lossy**.
- Lossy dimensionality reduction methods are Principal Components Analysis (PCA) and wavelet transforms.
- Principal Component Analysis (PCA) is to reduce the dimensionality of a data set by finding a new set of variables, smaller than the original set of variables, retains most of the sample's information and useful for the compression and classification of data.
- In PCA, it is assumed that the information is carried in the variance of the features, that is, the higher the variation in a feature, the more information that feature carries.

- Hence, PCA employs a linear transformation that is based on preserving the most variance in the data using the least number of dimensions.
- A Discrete Wavelet Transform (DWT) is a transform that decomposes a given signal into a number of sets, where each set is a time series of coefficients describing the time evolution of the signal in the corresponding frequency band.

4. **Numerosity reduction :** The numerosity reduction reduces the volume of the original data and represents it in a much smaller form. This technique includes two types parametric and non-parametric numerosity reduction.

- Log - linear models is an example of parametric method and nonparametric methods are histograms, clustering and sampling.
- Regression and log-linear linear regression models a relationship between the two attributes by modelling a linear equation to the data set.
- Log - linear regression analysis involves using a dependent variable measured by frequency counts with categorical or continuous independent predictor variables.

Clustering :

- Cluster analysis is a popular data discretization method. A clustering algorithm can be applied to discretize a numerical attribute (A) by partitioning the values of A into clusters or groups.
- Clustering takes the distribution of A into consideration, as well as the closeness of data points, and therefore is able to produce high-quality discretization results.
- Clustering can be used to generate a concept hierarchy for A by following either a top down splitting strategy or a bottom-up merging strategy, where each cluster forms a node of the concept hierarchy.

Sampling

- The basic idea of the sampling approach is to pick a random sample S of the given data D, and then search for frequent item sets in S instead of D.
- Sampling can be used as a data reduction technique because it allows a large data set to be represented by a much smaller random sample of the data. Different methods of sampling are Simple Random Sampling (SRS), stratified sampling, cluster sampling, systematic sampling and multistage sampling.

1.11 Data Discretization

Data Discretization :

- Data discretization means diving the range of continuous attribute into intervals. Actual data values are replaced by interval labels.

- It reduces the number of values for a given continuous attribute. Some classification algorithms only accept categorical attributes. It helps to a concise, easy-to-use, knowledge-level representation of mining results.
- Data discretization techniques can be categorized based on class information and which direction it proceeds. Class information is divided into two types : Supervised and unsupervised discretization. Categorized based on which direction it proceeds are of two types : Top - down and bottom - up.
- Discretization techniques can be classified as supervised and unsupervised discretization. Supervised discretization uses class information and unsupervised discretization does not use class information.
- **Top - down :** If the process starts by first finding one or a few points to split the entire attribute range, and then repeats this recursively on the resulting intervals. It is also called splitting.
- **Bottom - up :** It starts by considering all of the continuous values as potential split- points, removes some by merging neighbourhood values to form intervals, and then recursively applies this process to the resulting intervals. It is also called merging.
- Discretization can be performed recursively on an attribute to provide a hierarchical or multiresolution partitioning of the attribute values, known as a concept hierarchy. Concept hierarchies are useful for mining at multiple levels of abstraction.
- Discretization and concept hierarchy generation for numerical data uses following methods :

  a) Binning
  b) Histogram analysis
  c) Clustering analysis
  d) Entropy-based discretization
  e) Segmentation by natural partitioning

1.11.1 Concept Hierarchy Generation for Categorical Data

- Categorical data are discrete data. Categorical attributes have a finite number of distinct values, with no ordering among the values.
- Example : geographic location, job category and item type.
- Various methods are used for the generation of concept hierarchies for categorical data :

  a) Specification of a partial ordering of attributes explicitly at the schema level by users or experts

  - Example : A relational database or a dimension location of a data warehouse may contain the following group of attributes : Street, city, province or state and country.

- A user or expert can easily define a concept hierarchy by specifying ordering of the attributes at the schema level.
- A hierarchy can be defined by specifying the total ordering among these attributes at the schema level, such as : *street < city < province or state < country.*

b) Specification of a portion of a hierarchy by explicit data grouping

- We can easily specify explicit groupings for a small portion of intermediate - level data.
- For example, after specifying that area and country form a hierarchy at the schema level, a user could define some intermediate levels manually, such as : {India, Maharashtra, Pune} < SPPU.

c) Specification of a set of attributes, but not of their partial ordering

- A user may specify a set of attributes forming a concept hierarchy, but omit to explicitly state their partial ordering.
- The system can then try to automatically generate the attribute ordering so as to construct a meaningful concept.
- Example : Suppose a user selects a set of location-oriented attributes, street, country, state and city, from the database, but does not specify the hierarchical ordering among the attributes.
- Fig. 1.11.1 shows automatic generation of a schema concept hierarchy based on the number of distinct attribute values.

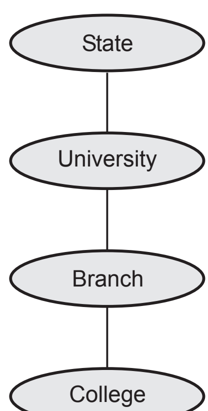

Fig. 1.11.1 Automatic generation of a schema concept hierarchy

1.12 Multiple Choice Questions

Q.1 Which of the following are NOT data reduction techniques ?

a. Data cube aggregation
b. Numerosity reduction
c. Attribute subset selection
d. Decimal scaling

Q.2 ______ normalization performs a linear transformation on the original data.

a. z-score
b. Min-max
c. Smoothing
d. Aggregation

Q.3 Which of the following methods to perform dimension reduction ?

a. Missing values
b. Decision tree
c. Random forest
d. All of these

Q.4 ______ is usually performed as an iterative two-step process consisting of discrepancy detection and data transformation.

a. Data reduction
b. Data integration
c. Data cleaning
d. Data transformation

Q.5 A _________ measure is a measure that must be computed on the entire data set as a whole.

a. distributive
b. holistic
c. unimodal
d. bimodal

Q.6 KDD stands for _________.

a. Knowledge Database in Discovery
b. Known Discovery in Databases
c. Known Distributed Databases
d. Knowledge Discovery in Databases

Q.7 Which of the following is NOT data preprocessing techniques ?

a. Data cleaning
b. Data integration
c. Data handling
d. Data transformation

Q.8 Concept hierarchies are a form of data _____ that can also be used for data smoothing.

a binaryzation
b discretization
c missing
d all of these

Q.9 Strategies for data reduction include _______.

a attribute subset selection
b numerosity reduction
c data cube aggregation
d all of these

Answer Keys for Multiple Choice Questions :

Q.1	d	**Q.2**	b	**Q.3**	d
Q.4	c	**Q.5**	b	**Q.6**	d
Q.7	c	**Q.8**	b	**Q.9**	d

❑❑❑